

Niraiva — Cancer-Related Clinical Dataset

1. Whole-Body (Systemic) Signs

These are general changes affecting the body as a whole that can be recorded as part of a patient's clinical history:

- **Unexplained weight loss:** A noticeable reduction in body weight without an intentional change in diet or exercise.
 - **Persistent fatigue:** Ongoing or unusual tiredness that does not improve normally with rest or sleep.
 - **Fever:** Elevated body temperature, including its duration and recurrence.
 - **Night sweats:** Episodes of excessive sweating occurring during sleep.
 - **Persistent pain:** Pain that continues over an extended period.
 - **Progressive pain:** Pain that increases in intensity, frequency, or affected area over time.
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2. Surface-Level Changes

These are observable changes involving the skin, body surface, or palpable areas:

- **New or changing lumps:** Newly detected lumps or areas of thickening, including their location, size, duration, and changes over time.
 - **Mole changes:** Changes in the size, shape, or color of an existing mole.
 - **Non-healing sores:** Sores or lesions that remain present instead of healing normally.
 - **Skin changes:** New or persistent changes in the appearance or condition of the skin.
 - **Skin yellowing (Jaundice):** Yellow discoloration of the skin or whites of the eyes.
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3. Internal & Functional Changes

These are changes involving normal body functions that can be recorded through patient history or clinical assessment:

- **Unusual bleeding:** Unexpected bleeding, including its location, duration, and frequency.
- **Blood in stool:** Presence of blood associated with bowel movements.

- **Blood in urine:** Presence of blood detected in urine.
 - **Coughing up blood:** Blood associated with coughing.
 - **Unexpected vaginal bleeding:** Vaginal bleeding occurring outside the individual's expected pattern.
 - **Persistent cough:** A cough that continues over an extended period.
 - **Hoarseness:** A persistent change in the normal quality of the voice.
 - **Difficulty swallowing:** Difficulty moving food or liquid through the swallowing process.
 - **Painful swallowing:** Pain or discomfort associated with swallowing.
 - **Bowel changes:** Persistent changes such as diarrhea, constipation, or changes in stool characteristics.
 - **Urinary changes:** Persistent changes in normal urination.
 - **Indigestion:** Persistent digestive discomfort.
 - **Difficulty eating:** Persistent difficulty with normal eating.
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4. Complete Blood Count (CBC) Triggers

The CBC provides measurements of blood-cell components that can be recorded as part of routine laboratory data:

- **Hemoglobin:** Records the measured hemoglobin concentration in the blood.
 - **Red Blood Cell (RBC) Count:** Records the number of red blood cells.
 - **White Blood Cell (WBC) Count:** Records the number of white blood cells.
 - **Platelet Count:** Records the number of platelets.
 - **RBC-related measurements:** Additional measurements associated with red blood-cell characteristics when available.
 - **WBC differential:** Records the distribution of different white blood-cell types when available.
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5. Comprehensive Metabolic Panel (CMP) Triggers

The CMP provides measurements relating to metabolic balance and organ function:

- **Alkaline Phosphatase (ALP):** Records the measured ALP concentration.

- **Aspartate Aminotransferase (AST):** Records the measured AST concentration.
 - **Alanine Aminotransferase (ALT):** Records the measured ALT concentration.
 - **Calcium:** Records the measured blood calcium concentration.
 - **Albumin:** Records the measured albumin concentration.
 - **Bilirubin:** Records the measured bilirubin concentration.
 - **Other metabolic measurements:** Additional relevant biochemical measurements when available.
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6. General Inflammatory Triggers

These are laboratory measurements that record inflammatory activity:

- **C-Reactive Protein (CRP):** Records the measured CRP concentration.
 - **Erythrocyte Sedimentation Rate (ESR):** Records the measured ESR.
 - **CRP/ESR history:** Records previous measurements when available to allow changes over time to be retained.
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7. Breast-Specific Signs and Changes

These variables record breast-related observations that can be obtained through patient history or clinical examination:

- **Breast lump:** Records the presence of a newly detected breast lump.
- **Breast thickening:** Records an area of unusual or localized thickening.
- **Breast location:** Records the anatomical location of a breast finding.
- **Lump size:** Records the measured dimensions of the breast finding.
- **Breast skin changes:** Records observable changes involving the breast skin.
- **Breast shape changes:** Records changes in breast appearance or contour.
- **Nipple changes:** Records changes involving the nipple.
- **Nipple discharge:** Records the presence and characteristics of nipple discharge.
- **Breast pain:** Records pain involving the breast.

- **Duration of breast finding:** Records how long the finding has been present.
 - **Progression of breast finding:** Records whether the finding has changed over time.
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8. Breast Imaging Data

These variables record observations obtained from breast imaging examinations:

- **Imaging method:** Records the type of imaging examination used.
 - **Lesion presence:** Records whether a lesion is identified.
 - **Lesion location:** Records the anatomical location of the lesion.
 - **Lesion size:** Records the measured size of the lesion.
 - **Lesion shape:** Records the observed shape of the lesion.
 - **Lesion margins:** Records the characteristics of the lesion margins.
 - **Number of lesions:** Records the number of identified lesions.
 - **Regional lymph-node findings:** Records observations involving regional lymph nodes.
 - **Distant findings:** Records observations concerning lesions or abnormalities outside the primary breast region.
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9. Pathology Data

These variables record information obtained from tissue or pathology examination:

- **Biopsy:** Records whether a tissue biopsy has been performed.
 - **Specimen type:** Records the type of tissue or specimen examined.
 - **Histological classification:** Records the observed pathological classification of the tissue.
 - **Tumor classification:** Records the pathological classification of the identified tumor.
 - **Pathology observations:** Records relevant findings reported from the tissue examination.
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10. HER2 Data

These variables record HER2-related laboratory and pathology measurements:

- **HER2 Immunohistochemistry (IHC):** Records the HER2 staining result obtained through immunohistochemical testing.
 - **HER2 FISH:** Records the result obtained through fluorescence in situ hybridization testing.
 - **HER2 status:** Records the reported HER2 status.
 - **ERBB2 gene information:** Records available information concerning ERBB2 gene amplification or duplication.
 - **HER2 ratio:** Records the ratio obtained from the relevant HER2 laboratory assessment.
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11. Estrogen Receptor (ER) Data

These variables record estrogen-receptor information obtained from tissue testing:

- **ER immunostaining:** Records the observed estrogen-receptor staining.
 - **ER status:** Records the reported ER status.
 - **ER percentage:** Records the percentage of tested cells showing ER staining when provided.
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12. Progesterone Receptor (PR) Data

These variables record progesterone-receptor information obtained from tissue testing:

- **PR immunostaining:** Records the observed progesterone-receptor staining.
 - **PR status:** Records the reported PR status.
 - **PR percentage:** Records the percentage of tested cells showing PR staining when provided.
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13. TNM Classification Data

These variables record the individual components used to describe the anatomical extent of a tumor:

T — Primary Tumor

- **T category:** Records the assigned T classification.

- **Primary tumor size:** Records the measured size of the primary tumor.
- **Local tumor extension:** Records documented information concerning extension of the primary tumor.

N — Regional Lymph Nodes

- **N category:** Records the assigned N classification.
- **Number of involved lymph nodes:** Records the number of regional lymph nodes involved.
- **Micrometastasis:** Records documented microscopic metastatic involvement.
- **Macrometastasis:** Records documented macroscopic metastatic involvement.

M — Distant Metastasis

- **M category:** Records the assigned M classification.
 - **Distant metastasis:** Records documented distant metastatic involvement.
 - **Metastasis location:** Records the documented anatomical location of distant metastatic findings.
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14. Patient & Observation Metadata

These fields provide context for every individual piece of data stored in the dataset:

- **Patient age:** Records the patient's age at the time of observation.
- **Observation date:** Records when the observation or test was performed.
- **Data source:** Records where the information originated, such as patient history, physical examination, laboratory test, imaging, or pathology.
- **Measurement value:** Stores the actual numerical or categorical value.
- **Unit:** Records the measurement unit when applicable.
- **Reference range:** Records the relevant laboratory reference range when applicable.
- **Previous measurement:** Stores earlier values when available.
- **Current measurement:** Stores the most recent available value.
- **Observation duration:** Records how long a symptom or physical finding has been present when applicable.
- **Observation progression:** Records whether a finding has changed over time.

Lung Cancer Probability Dataset Based on Gender and Smoking History

1. Gender

The dataset first separates individuals according to the recorded gender variable:

- **Female:** Gender recorded as female.
- **Male:** Gender recorded as male.

2. Smoking Status

The dataset records the individual's smoking status:

- **Smoker:** Currently recorded as a smoker.
- **Quit Smoker:** Previously smoked and has stopped smoking.
- **Nonsmoker:** Recorded as a nonsmoker.

3. Smoking Cessation Age

For individuals who have quit smoking, the dataset considers the age at which smoking was stopped:

- **Quit-smoking age:** Records the age at which the individual stopped smoking.
- **Quit-smoking age not available:** Records cases where the relevant cessation-age information is not available.

4. Lung Cancer Probability

This variable stores the probability of lung cancer associated with the corresponding combination of gender and smoking-related characteristics in the dataset:

- **Female smoker:** 0.185 — 18.5%
- **Female quit smoker:** 0.026 — 2.6%
- **Female nonsmoker:** 0.004 — 0.4%
- **Male smoker:** 0.244 — 24.4%
- **Male quit smoker:** 0.055 — 5.5%
- **Male nonsmoker:** The exact value is not fully visible in the supplied tree, so it cannot be reliably extracted